SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation **COURSE CODE: CSA13**

COURSE OUTCOME COVERED

CO1: *Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems. (BL:3)*

Assessment Tool Weightage: 20%

QUESTIONS: 2 **Total Marks:50**
Duration : 1 Hour

Simulation

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 (25)	Q2 (25)	Total Marks (50)
Conceptual Understanding	30%	7.5	7.5	15
Accuracy of Answer	25%	6.25	6.25	12.5
Application of Knowledge	20%	5	5	10
Completeness of Response	15%	3.75	3.75	7.5
Clarity & Presentation	10%	2.5	2.5	5
Total per Question	**100%**	**25**	**25**	**50**

1 Design a **Non-Deterministic Finite Automaton (NBA)** for an **Automatic Door System** based on sensor inputs.An automatic door system can be modeled using an NBA to represent different states and transitions. The system consists of four states: *Closed, Opening, Open,* and *Closing*. The input symbols are P (Person detected) and N (No person detected). Initially, the door is in the *Closed* state. When a person is detected (P), the system may non-deterministically transition from *Closed* to *Opening*. Once opened, the system moves to the *Open* state. If no person is detected (N), the door transitions to *Closing* and eventually returns to the *Closed* state. Non-determinism occurs when the system may remain in the *Open* state even if a person is detected or may start closing if no detection occurs.

2 Design a finite automaton to model a traffic light control system at a road intersection. The automaton should represent different states corresponding to the traffic lights, such as Red, Green, and Yellow. Define the transitions between these states based on a timer or control signals, ensuring that the sequence follows the standard order (Red → Green → Yellow → Red). Include considerations for timing constraints, such as how long each light remains active, and describe how the automaton handles continuous cycling of signals. Additionally, explain how this model can be extended to handle pedestrian crossing signals or emergency vehicle overrides, ensuring safe and efficient traffic flow.

RUBRICS FOR CALCULATION

